U1 Exploring One-Variable Data and Collecting Data

1.0 Introduction Statistics

Observational Unit — An object described in a set of data
- can be people, animals, or things.
- Individual: mostly refer to people.

Categorical (Qualitative) Variable — Assigns labels that place each individual into a particular group, called a category

Quantitative Variable — Takes number values that are quantities — counts or measurements.

Identifier Variable — Categorical variable with exactly one individual in each category.

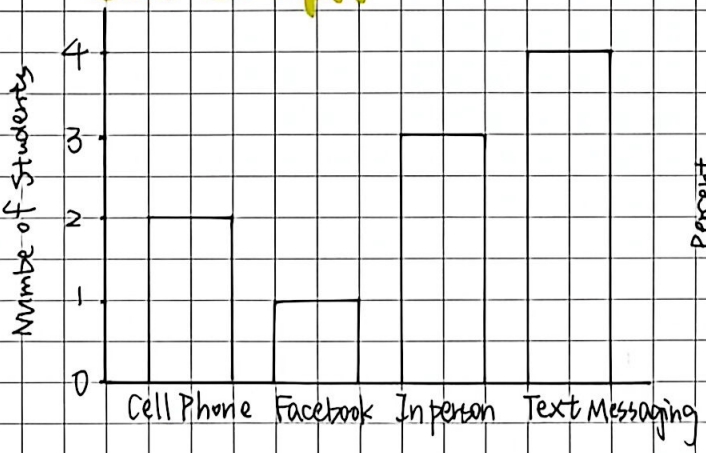
1.1 Analyzing Categorical Data

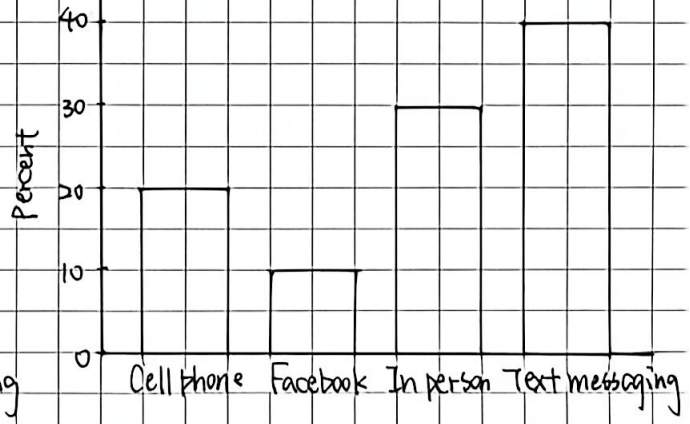
Frequency Table

Frequency Table	
Preferred method	Frequency
Cell phone	2
Facebook	1
In person	3
Text messaging	4

Relative Frequency Table	
Preferred method	Relative Frequency
Cell phone	2/10 = 0.2 or 20%
Facebook	1/10 = 0.1 or 10%
In person	3/10 = 0.3 or 30%
Text messaging	4/10 = 0.4 or 40%

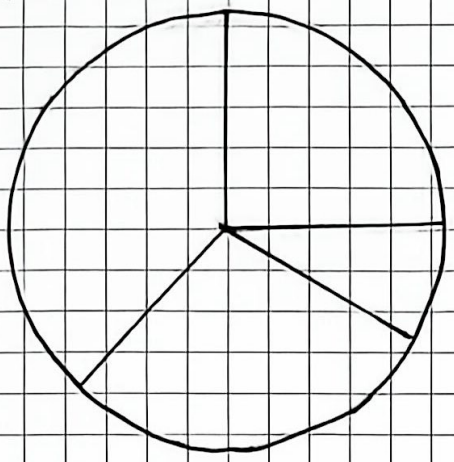
Bar Graph





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Pie Chart

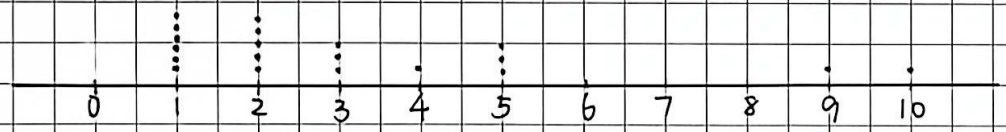


1.2 Analyzing Quantitative Data

Dotplot

5	5	1	10	5	2	1	1	2	3
3	2	1	4	2	1	2	1	9	3

→ Data → Quick Graph



Stemplot

F:	50	26	26	31	57	19	24	22	23	38
M:	13	50	13	34	23	30	49	13	15	51

1	3 3 3 3 9
2	2 3 3 4 6 6
3	0 1 4 8
4	9
5	0 0 1 7

Key: 4|9 represents a male student who reported having 49 pairs of shoes

Back-to-back Stemplot

	9 | 1	3 3 3 5
6 6 4 3 2	2	3
8 1	3	0 4
	4	9
7 0	5	0 1

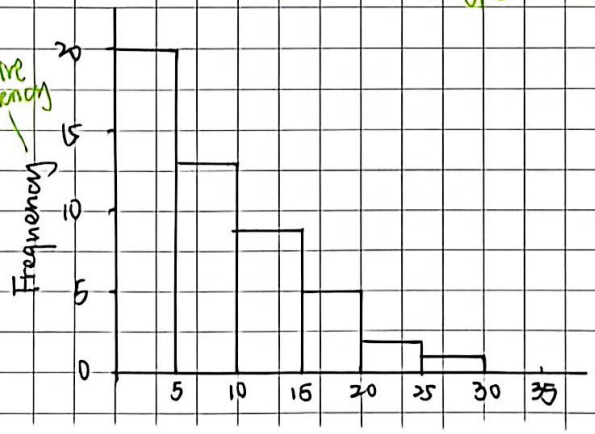
Key: 4|9 represents a student who reported having 49 pairs of shoes

Histogram

Menu → Plot type → Histogram

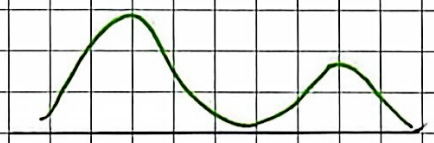
Frequency Table	
Class	Count
$0 \le x < 5$	20
$5 \le x < 10$	13
$10 \le x < 15$	9
$15 \le x < 20$	5
$20 \le x < 25$	2
$25 \le x < 30$	1
Total	50

Or relative frequency



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Bimodal — · two clear peaks
· two centers
· Gap



Uniform — · each bar height is approximately the same
· no prominent peaks



Describing Distribution

- Shape { Unimodal { symmetric / skew to left / skew to right
  Bimodal
  Uniform
- Center: Mean / median
- Variability: Range / IQR
- Unusual features { Gaps / Clusters
  Outliers

Comparing Distributions

- Use comparative words: same / different / more / less

1.4 Describing Location in a Distribution

Percentiles — · The pth percentile is interpreted as the value that has p% of the data less than or equal to it

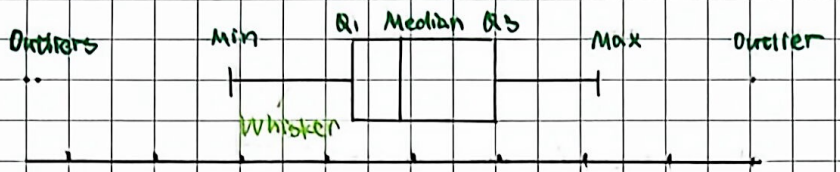
z-scores (standardized scores)

Tells how many standard deviations from the mean an observation falls, and in what direction

- $z = \frac{x - \mu}{\sigma}$

…'s … is … standard deviations above / below the mean of …

Boxplot → data → quick graph → menu → plot type → boxplot



Five number Summary — min, Q_1, median, Q_3, max

1.3 Describing Quantitative Data with Numbers

Measuring Center

Mean — · $\bar{x}$ → menu → statistics → calculation → one-variable → $\bar{x}$
- sensitive to extreme

Median — · resistant to extreme

Measuring Variability

Range — · Range = Maximum − Minimum ★ A single number
sensitive to extreme

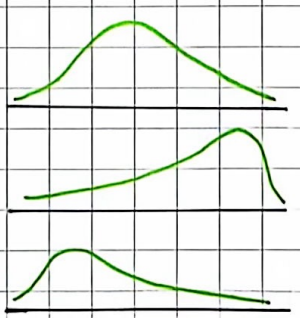
Standard Deviation — sensitive to extreme
· the typical distance of the values in a distribution of ... from the mean ...

IQR (Interquartile range) — · IQR = $Q_3 - Q_1$
- Q_1 左四分位
- Q_3 右四分位

★ **Identifying Outliers: The 1.5 × IQR Rule**
- lower outliers < $Q_1 - 1.5 \times IQR$
- Higher outliers > $Q_3 + 1.5 \times IQR$

Describing Shape

Unimodal — · single peak
- Symmetric — mean ≈ median
- Skew to left — mean < median
- Skew to right — mean > median



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1.5 Sampling and Surveys

Population – the entire group of individuals we want information about.

Statistic vs Parameter

Name	Statistic (sample)	Parameter (population)
Mean	$\bar{x}$	μ
Standard Deviation	s	σ
Correlation	r	ρ
Regression Coefficient	b_1	β_1
Proportion	$\hat{p}$	p

Sampling Methods

Simple Random Sample (SRS)

every group of a given size has an equal chance of being chosen

How? label: give each individual in the population a distinct numerical label from 1 to N

randomize: use a random number generator to obtain n different integers from 1 to n

select: choose the individuals that correspond to the randomly selected integers

Stratified Random Sample

the division of population into separate groups, called **strata**, based on shared attribute or characteristics (homogeneous grouping)

Cluster Random Sample

the division of a population into smaller groups, called **cluster**, each cluster similar to each other (heterogeneous grouping)

Systematic Random Sample

Sample members from a population are selected according to a random starting point and a fixed, periodic interval i.e. ⑪ 共 ⑯ 共 ㉑ 共 ㉖

Census

Selects all items/subjects in a population

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Sampling Bias — · Bias
· Sampling Variability

Voluntary Response Bias
When a sample is compromised entirely of volunteers or people who choose to participate, the sample will typically not be representative of the population.
· Only people who will participate, so it will over/under estimate the result.

Undercoverage Bias
· When part of the population has a reduced chance of being in the sample, ...

Nonresponse Bias
· Individuals chosen for the sample for whom data cannot be obtained (or who refuse to respond) may differ from those for whom data can be obtained

Response Bias
Problems in the data gathering instrument or process result in response bias.
i.e misleading framing of questions

Convenience Sample Bias
Non-random sampling methods introduce potential for bias because they do not use chance to select the individual

1.6 Experiments

Observational Study — Treatments are NOT imposed
· Retrospective: examine data for a sample of individuals (已有)
· Prospective: follow a sample of individuals into the future collecting data (未来)

Experiment — Treatments are imposed
· manipulate factor levels
· compare responses

Treatment — the levels or combination of levels of the explantory variable(s) are called treatments

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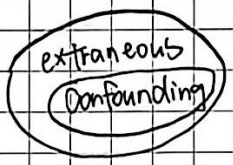
Experimental units — the individuals (which may be people or other objects of study)

Explanatory variable = Independent variable — A variable whose levels are manipulated intentionally

Response Variable = dependent variable — The outcome from the experimental units that is measured after the treatments have been administered.

Extraneous Variable — Known / of believed to affect the response but is not an explanatory variable

Confounding Variable — A variable that is related to the explanatory variable and influences the response variable



Other Terminology

Blinding — Any individual associated with an experiment who is not aware of how subjects have been allocated to treatment groups
- Single-blind: Subjects do not know, researchers know
- Double-blind: Subjects & researchers don't know

Placebo — A treatment known to have no effect

Placebo Effect — Occurs when experimental units have a response to a placebo.

Control Group — A collection of experimental units either not given a treatment of interest or given a treatment with placebo

4 Principles of a Well-designed Experiment

Comparison — comparisons of at least 2 experiment

Random Assignment — Equalize the effects of unknown / uncontrollable sources of variation.
- allow for causation conclusions

Replication — enough experimental units (multiple for each treatment)

Control — making conditions as similar as possible to control potential confounding variables.

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Experimental Design

Completely Randomized Design

- Treatments are assigned to experimental units completely at random

Randomized (Complete) Block Design

- Blocking ensures that at the beginning of the experiment within each block are similar to each other with respect to at least one variable.
- Help to separate natural variability from differences due to the blocking variable
- Blocking separate the variation in the response to allow for a more precise comparison of the treatments.

Matched Pairs Design

- Special case of randomized block design.
- Using a blocking variable, subjects are arranged in pairs matched on relevant factors.
- Each pair receives both treatments by randomly assigning one treatment to one member of the pair

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U2 Probability, Random Variables, and Probability Distributions

2.1 Analyzing Data on Two Categorical Variables

Two-Way tables (Contingency Table)

- A table of counts that summarizes data on the relationship between two categorical variables for some group of individuals

		Environmental Club		
		No	Yes	Total
Snowmobile Use	Never Used	445	212	657
	Snowmobile renter	497	77	574
	Snowmobile Owner	279	16	295
	Total	1221	305	1526

Marginal Frequency — row and column totals divided by the total for the entire table

- $\frac{\text{Row (Column) Total}}{\text{Total}}$ i.e. $\frac{657}{1526}$

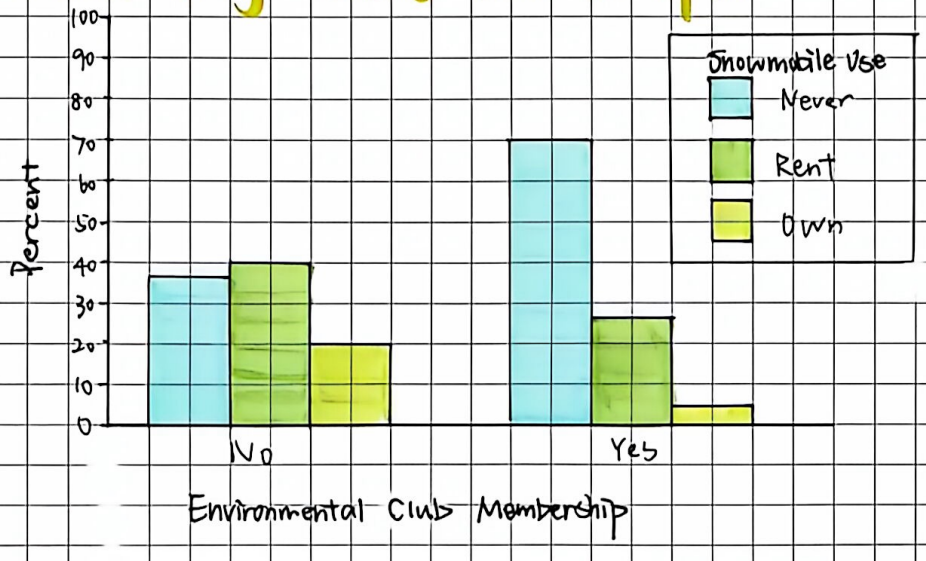
Joint Relative Frequency — a cell frequency divided by the total for the entire table

- i.e. Yes + Owner = $\frac{16}{1526}$

Conditional Relative Frequency — a relative frequency for a specific part of the table

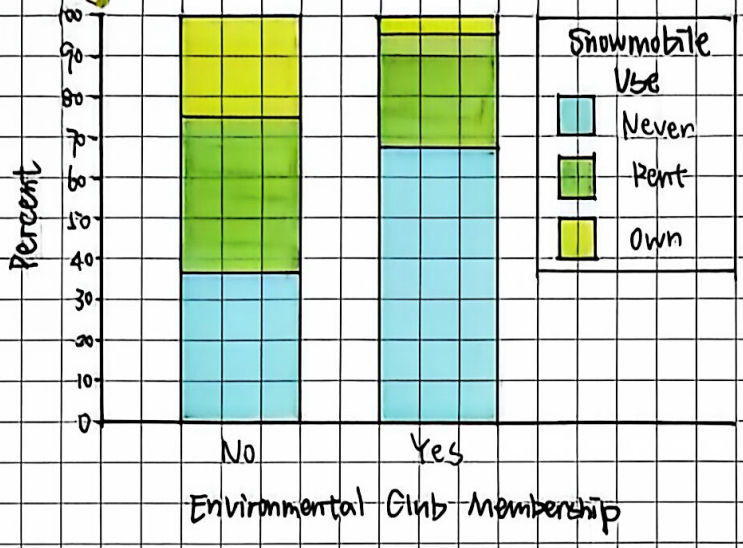
- i.e. Owner | Yes = $\frac{16}{305}$

Side-by-side Bar Graph

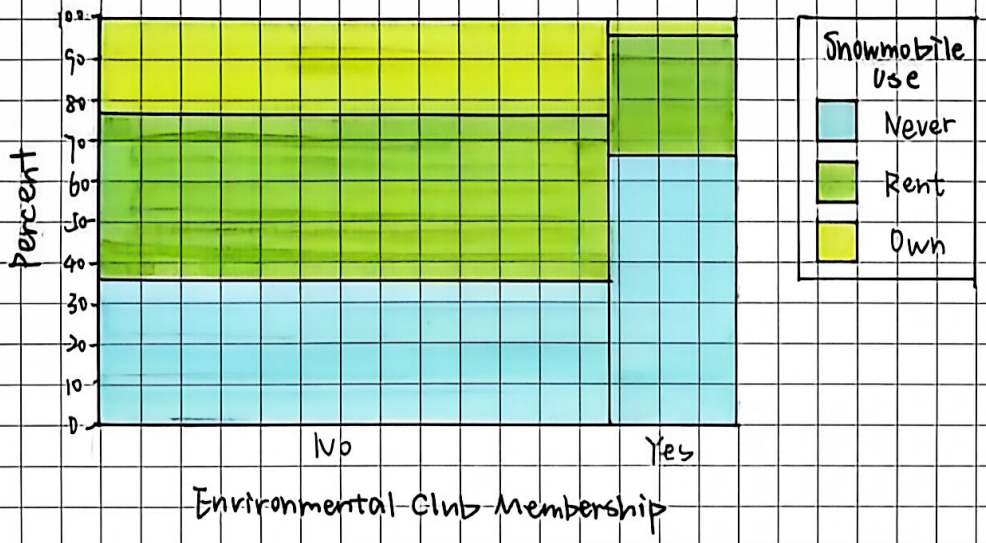


Segmented Bar Graph

Displays the same information as a pie chart



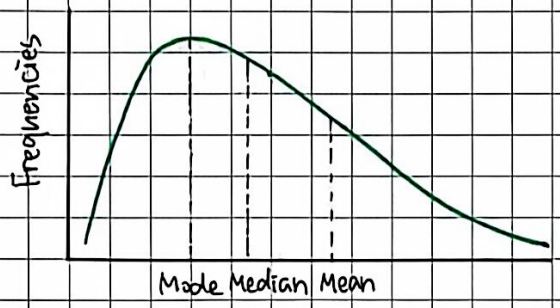
Mosaic Plot



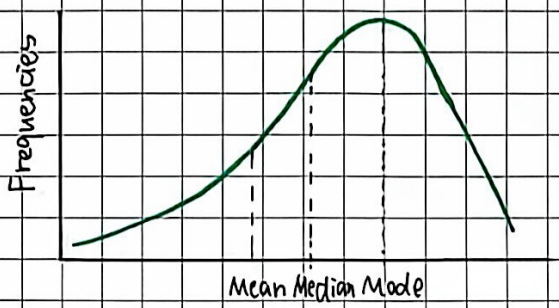
2.2 Density Curves and Normal Distributions

Density Curves

- is always on or above the horizontal axis
- has area exactly 1 underneath it



A positively skewed distribution

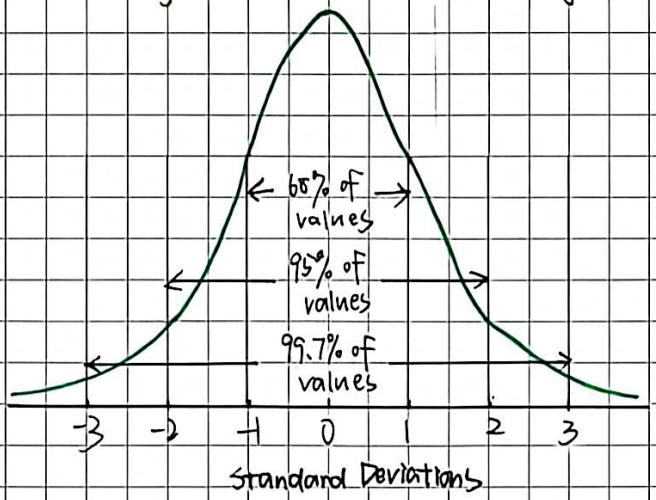


A negatively skewed distribution

Mean — μ Standard Deviation — σ

Normal Distributions

- Shape: symmetric, single-peaked, bell-shaped
- Center: μ = median, located at the midpoint of the symmetric curve
- Variability: σ measures the variability (width) of a normal distribution



The 68-95-99.7 Rule

- In a Normal distribution with mean μ and standard deviation σ

Approximately 68% of the observations fall within σ of the mean μ
Approximately 95% of the observations fall within 2σ of the mean μ
Approximately 99.7% of the observations fall within 3σ of the mean μ

Standard Normal Distribution

- The normal distribution having a mean of 0 and a standard deviation of 1
- $z = \frac{x - \mu}{\sigma}$

↳ ① 一个 observation 距离 mean 有几个 SD ② tells distance & direction

Finding Areas in a Normal Distribution

Area = Calculate → menu → stats → Distribution → normalCDF(lower, upper, μ, σ)
Bound (x) ... → InverseNorm (Area, μ, σ)

Normality

Normal Probability plot: If the points on a Normal Probability Plot lie close to a straight line, the data are approximately NORMALLY distributed



approximately Normal　　not Normal　　approximately Normal (outlier)

2.3 Probability

Permutation & Combination

Factorial – For any positive integer n, $n! = n(n-1)(n-2)\cdots 2\cdot 1$
- $0! = 1$

Permutation 排列 – the number of ways a set of elements can be ordered.
- $_nP_r = \frac{n!}{(n-r)!} = A^r_n$

Combination 组合 – the number of combinations of r elements chosen from a set of n elements
- $_nC_r = \frac{_nP_r}{r!} = \frac{n!}{r!(n-r)!} = C^r_n$

Probability

- Definition: long-run proportion of times an outcome/event A occurs P(A)
- range: $0 \le P(A) \le 1$
  - $P(A) = 0$ impossible
  - $P(A) = 1$ certain

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Simulation

- Definition: using a random process to model real-world random events
- Trial = one complete repetition of the simulation
- Outcome = result of a trial
- Empirical probability = probability estimated from simulation / data
- $\hat{P}(A) = \frac{\text{\# trials in which A occurs}}{\text{\# total trials}}$
- Process =
  - Assign numbers to possible outcomes according to their probabilities
  - Generate random numbers to imitate the chance process
  - Record the result of one complete trial
  - Repeat many trials
  - Calculate the proportion of trials where the event occurs

Law of Large Numbers

- As the number of trials increases, empirical probability tends to get closer to the true probability

2.4 Probability Rules

Probability Model

A description of some chance process: a list of possible outcomes + probability for each outcome

Sample Space

- List of all possible outcomes
- $S = \{a, b, c\}$

Event

Event ⊂ Sample Space

- A collection of outcomes

Equally Likely Outcomes

- If all outcomes in the sample space are equally likely,

$$P(A) = \frac{\text{\# of outcomes in event A}}{\text{total \# of outcomes in sample space}}$$

Complement Rule

- Complement of event A = A^c
- $P(A^c) = 1 - P(A)$

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Disjoint Events (Mutually Exclusive)

Two events have no outcomes in common and so can never occur together

- $P(A \text{ and } B) = 0$ / $P(A \cap B) = 0$

Additive Rule for Mutually Exclusive Events

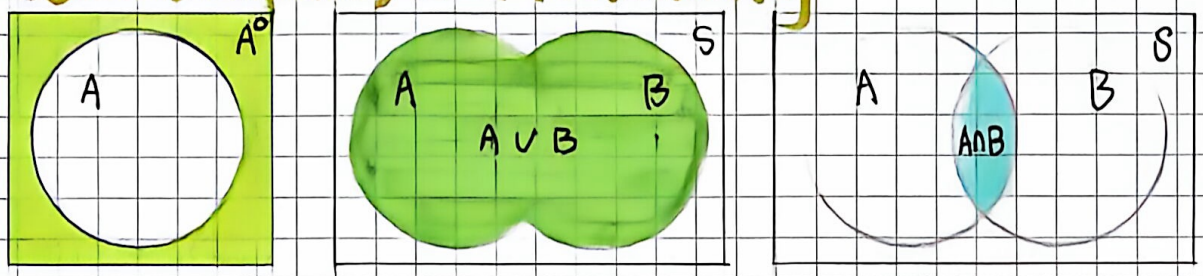
- $P(A \text{ or } B) = P(A) + P(B)$ / $P(A \cup B) = P(A) + P(B)$

General Additional Rule

($\cup$ = or (union)
$\cap$ = and (intersection)

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Venn Diagrams and Probability



Complement A^C | Union $A \cup B$ | Intersect $A \cap B$

(≠ mutually exclusive)

2.5 Conditional Probability and Independence

Conditional Probability

- The probability that one event happens given that another event is known to have happened
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$

General Multiplication Rule

- $P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$

Independent Events

- whether or not one event has occurred does not change the probability that the other event will happen
- $P(A|B) = P(A|B^c) = P(A)$
  $P(B|A) = P(B|A^c) = P(B)$

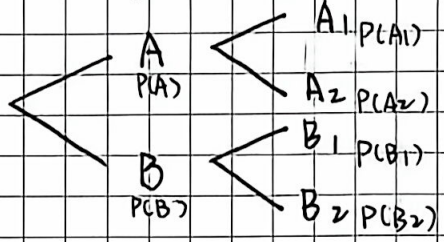
Multiplication Rule for Independent Events

- $P(A \cap B) = P(A) \cdot P(B)$

* with replacement = independent, vs without replacement = dependent

Tree Diagram

- shows the sample space of a chance process involving multiple stages



2.6 Random Variables

Random Variable

- numerical value describing the outcome of a chance process
- X
- * define X = number of ...

Discrete Random Variable

- countable values
- usually comes from counting

Probability Distribution

- lists all possible values of X and their probabilities
- $0 \leq p_i \leq 1$
- $\Sigma p_i = 1$

Mean / Expected Value

- represents the long-run average value of X
- E(X) does NOT have to be a possible value of X
- $\mu_X = E(X) = \Sigma x_i p_i$

⊞ → x_i list = 'value'
frequency list = 'probabilities'

Standard Deviation of a Random Variable

- measures the typical distance of X from its mean
- larger σ_X → more variability
- $\sigma_X = \sqrt{\Sigma (x_i - \mu_X)^2 p_i}$

Continuous Random Variables

- can take any value in an interval
- usually comes from measuring
- probability = area under the density curve

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2.7 Binomial Distribution

Binomial Setting – BINS

A setting is binomial if:

- **B** – **Binary**: success / failure
- **I** – **Independent**: trials are independent
- **N** – **Number**: fixed number of trials n in advance 提前定固定数
- **S** – **Same probability**: same p for every trial (success probability)

- Define: X = number of successes
  possible values: $X = 0, 1, 2, \ldots, n$

Binomial Probability Formula

- $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

  - n = number of trials
  - p = probability of success
  - k = number of successes
  - $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ = ways to arrange k successes

binompdf$(n, p, k) = P(X=k)$ exactly
binomcdf$(n, p, k) = P(X \le k)$ cumulative
$1 - $ binomcdf$(n, p, k-1) = P(X \ge k)$

Mean & SD of a Binomial Random Variable

- Mean: $\mu_X = np$
  - In many repetitions, the average number of successes will be about ...
- SD: $\sigma_X = \sqrt{np(1-p)}$
  The number of success typically differs from the mean by about ...

Normal Approximation to Binomial

10% Condition

- When sampling WITHOUT replacement from a population of size N:

  $n \le 0.10N$

  → trials can be treated as approximately independent
  → allows us to use a Binomial model

Large Counts Condition

- The binomial distribution can be approximated by a Normal distribution if:

  $np \ge 10$ and $n(1-p) \ge 10$

  $np \ge 10$: expected number of success
  $n(1-p) \ge 10$: expected number of failures

Normal Approximation

- If Large Counts is satisfied: $X \sim N(np, \sqrt{np(1-p)})$
- Therefore: $\mu = np$, $\sigma = \sqrt{np(1-p)}$ ⇒ normalcdf

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